

CLASS 10 · ICSE · SELINA CONCISE MATHEMATICS · PYQ SOLUTIONS

Chapter 25: Probability

Step-by-step solutions to the previous year questions

The question numbers match the PYQ question sheet.

1 Probability of an event

1. Which of these cannot be the probability of an event? (a) $\frac{5}{4}$ (b) 0.25 (c) $\frac{1}{33}$ (d) 67%

ICSE 2025, 1 mark

SOLUTION

A probability always lies from 0 to 1. $\frac{5}{4} = 1.25$ is greater than 1.

Answer: (a)

2 Balls, numbered cards and letters

2. A box holds cards numbered 1 to 30, and one card is drawn at random. A wall clock is won for a perfect square, a water bottle for an even number that is also a multiple of 3, and a purse for a prime number. Find the probability of winning (a) the wall clock (b) the water bottle (c) the purse.

ICSE 2025, 3 marks

SOLUTION

There are 30 equally likely outcomes.

(a) Perfect squares: 1, 4, 9, 16, 25 (5 cards). Probability = $\frac{5}{30} = \frac{1}{6}$.

(b) Even multiples of 3 are the multiples of 6: 6, 12, 18, 24, 30 (5 cards). Probability = $\frac{5}{30} = \frac{1}{6}$.

(c) Primes: 2, 3, 5, 7, 11, 13, 17, 19, 23, 29 (10 cards). Probability = $\frac{10}{30} = \frac{1}{3}$.

Answer: (a) $\frac{1}{6}$ (b) $\frac{1}{6}$ (c) $\frac{1}{3}$

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